CRASHING OF PROJECT NETWORK

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(1) Consider the data of a project as shown in the following table:

Activity	Normal Time (weeks)	Normal Cost (Rs.)	Crash time (weeks)	Crash Cost (Rs.)
1-2	7	700	4	850
1-3	5	500	3	700
1-4	8	600	5	1,200
2-5	9	800	7	1,250
3-5	5	700	3	1,000
3-6	6	1,100	5	1,300
4-6	7	1,200	5	1,450
5-7	2	400	1	500
6-7	3	500	2	850

If the indirect cost per week is Rs. 200, find the optimal crashed project completion time.

$$\text{Slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal Time} - \text{Crash Time}}$$

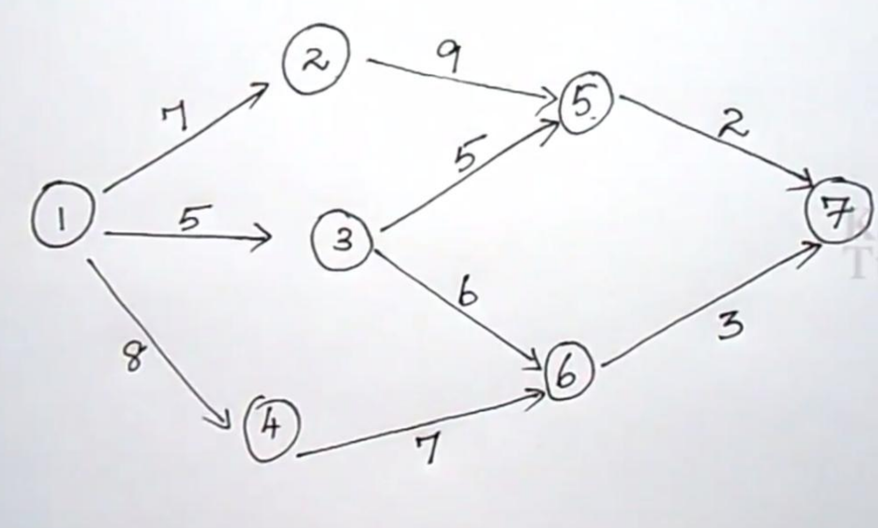
Activity	Normal Time (Weeks)	Normal Cost (Rs.)	Crash time (Weeks)	Crash Cost (Rs.)	Slope
1-2	7	700	4	850	

$$\text{Slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal Time} - \text{Crash Time}}$$

$$= \frac{850 - 700}{7 - 4} = \frac{150}{3} = 50$$

Activity	Normal Time (Weeks)	Normal Cost (Rs.)	Crash time (Weeks)	Crash Cost (Rs.)	Slope
1-2	7	700	4	850	50
1-3	5	500	3	700	100
1-4	8	600	5	1,200	200
2-5	9	800	7	1,250	225
3-5	5	700	3	1,000	150
3-6	6	1,100	5	1,300	200
4-6	7	1,200	5	1,450	125
5-7	2	400	1	500	100
6-7	3	500	2	850	350

I Iteration



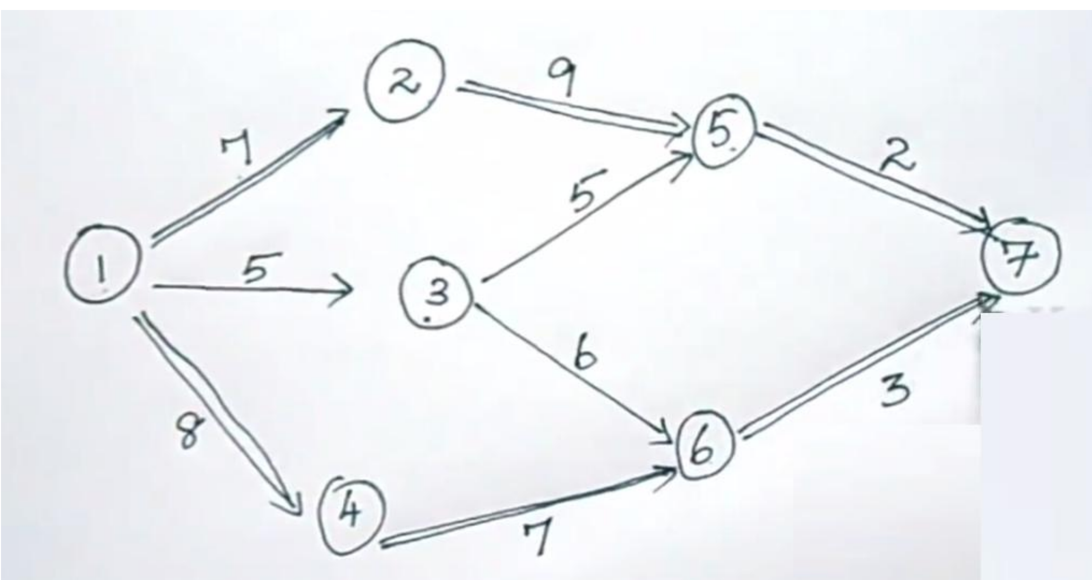
Activity	Normal Time(Week
1-2	7
1-3	5
1-4	8
2-5	9
3-5	5
3-6	6
4-6	7
5-7	2
6-7	3

1-2-5-7 = 7+9+2 = 18 Critical path

1-3-5-7 = 5+5+2 = 12

1-3-6-7 = 5+6+3 = 14

1-4-6-7 = 8+7+3 = 18 Critical path



Activity	Normal Time (weeks)	Normal Cost (Rs.)	Crash time (Weeks)	Crash Cost (Rs.)	Slope
1-2	7	700	4	850	50
1-3	5	500	3	700	100
1-4	8	600	5	1,200	200
2-5	9	800	7	1,250	225
3-5	5	700	3	1,000	150
3-6	6	1,100	5	1,300	200
4-6	7	1,200	5	1,450	125
5-7	2	400	1	500	100
6-7	3	500	2	850	350
		6,500			

Normal Porject Comp. time, 18weeks

Critical path.

1-2-5-7

and

1-4-6-7

Total Direct normal cost, 6,500

Indirect cost (200 x 18) 3,600

10,100

Crash Limit and Slope

Critical path	Critical activity	Crash Limit	Cost Slope
1-2-5-7	1-2	3	50
	2-5	2	225
	5-7	1	100
1-4-6-7	1-4	3	200
	4-6	2	125
	6-7	1	350

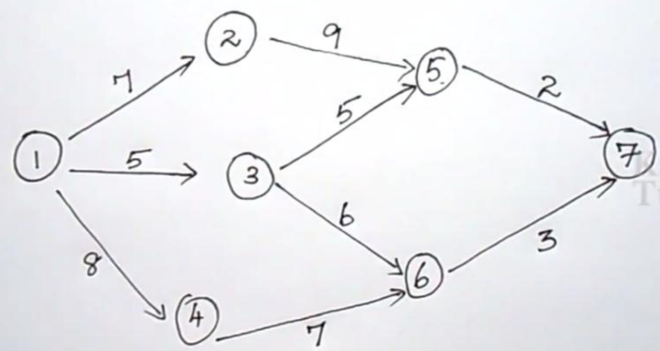
Crash Limit = Normal time – Crash time

Activity	Normal Time (Weeks)	Normal Cost (Rs.)	Crash time (Weeks)	Crash Cost (Rs.)	Slope
1-2	7	700	4	850	50
1-3	5	500	3	700	100
1-4	8	600	5	1,200	200
2-5	9	800	7	1,250	225
3-5	5	700	3	1,000	150
3-6	6	1,100	5	1,300	200
4-6	7	1,200	5	1,450	125
5-7	2	400	1	500	100
6-7	3	500	2	850	350
		6,500			

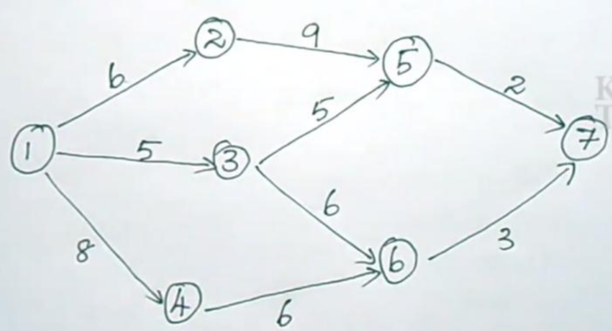
Crash Limit and Slope

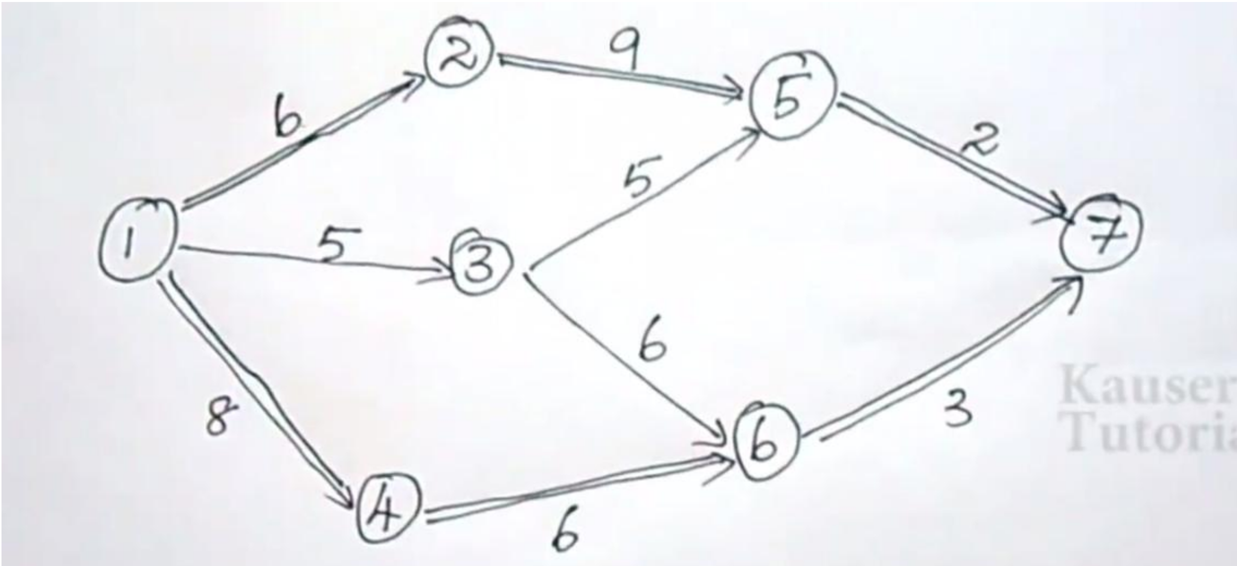
Critical path	Critical activity	Crash Limit	Cost Slope
1-2-5-7	1-2	3	50 *
	2-5	2	225
	5-7	1	100
1-4-6-7	1-4	3	200
	4-6	2	125 *
	6-7	1	350

I Iteration



II Iteration





✓ 1-2-5-7 = 6+9+2 = 17

1-3-5-7 = 5+5+2 = 12

1-3-6-7 = 5+6+3 = 14

✓ 1-4-6-7 = 8+6+3 = 17

T.C= pre.total cost + direct cost(slope cost) - indirect cost

Proj. comp. time = 17 weeks

1-2-5-7

and

1-4-6-7

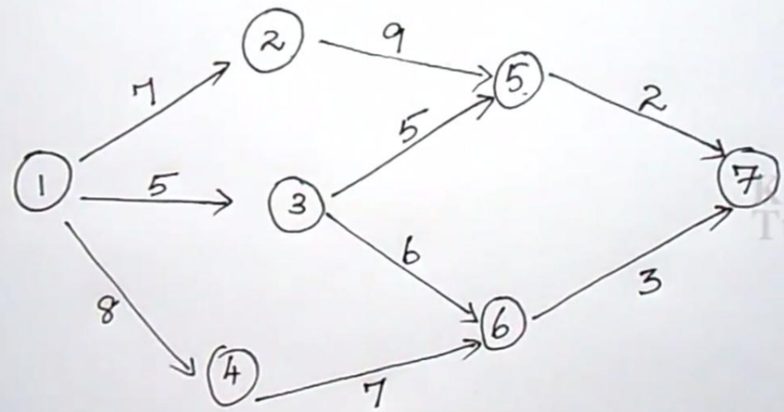
10,100 + [50+125] − 200

T. cost = 10,075

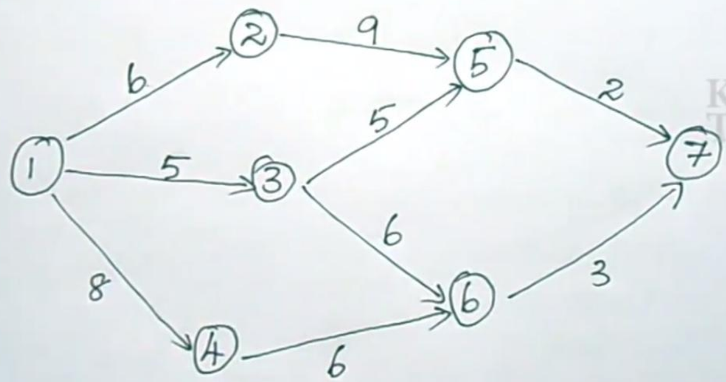
Crash Limit and Slope.

Critical path	Critical activity	Crash Limit	Cost Slope.
1-2-5-7	1-2	2	50*
	2-5	2	225
	5-7	1	100
1-4-6-7	1-4	3	200
	4-6	1	125*
	6-7	1	350

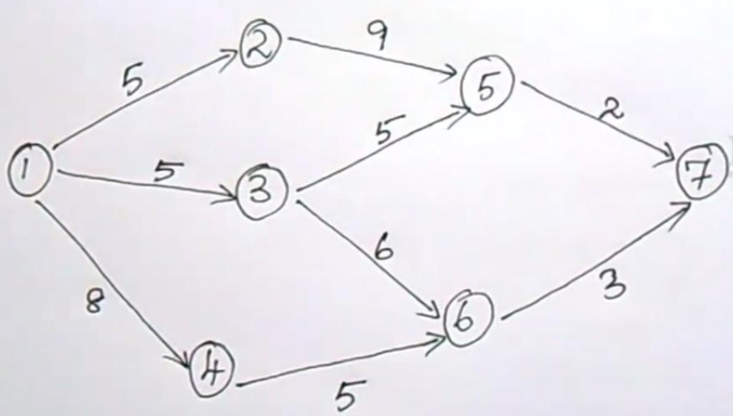
I Iteration

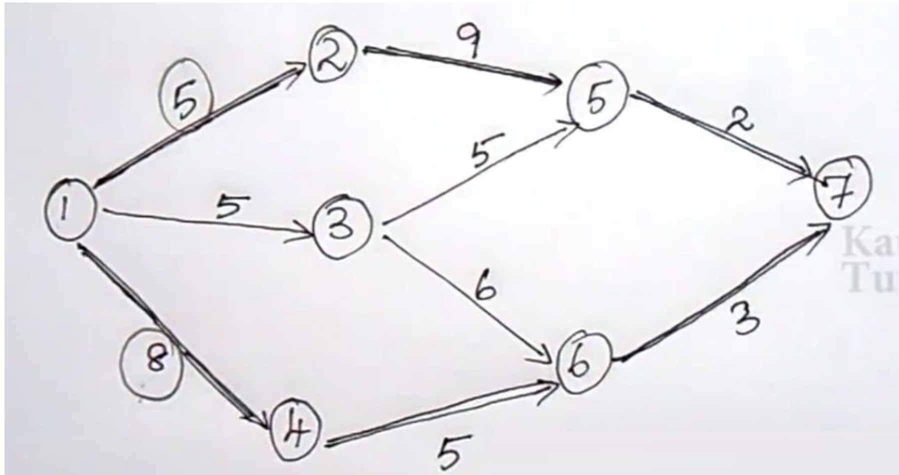


II Iteration



III Iteration





5 Proj. comp. time = 16 weeks

✓ 1-2-5-7 = 5+9+2 = 16

1-3-5-7 = 5+5+2 = 12

1-3-6-7 = 5+6+3 = 14

✓ 1-4-6-7 = 8+5+3 = 16

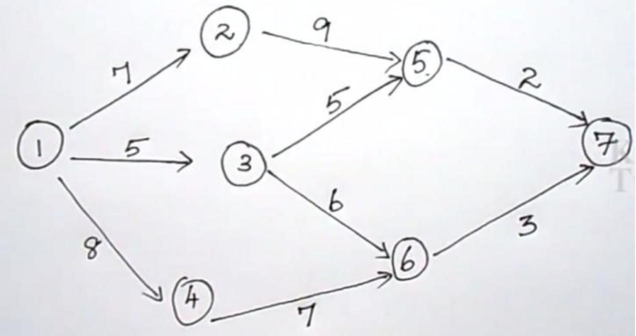
$10{,}075 + [50 + 125] - 200$

$= 10{,}050$

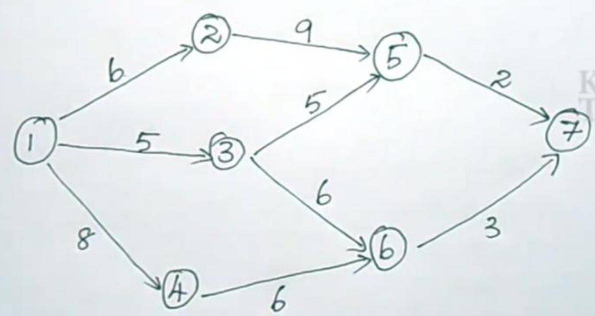
Crash Limit and Slope

Critical path	Critical activity	Crash Limit	Cost Slope
1-2-5-7	1-2	1	50 *
	2-5	2	225
	5-7	1	100
1-4-6-7	1-4	3	200 *
	4-6	0	125
	6-7	1	350

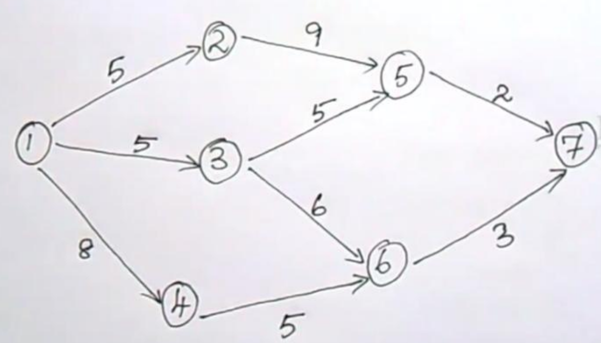
I Iteration



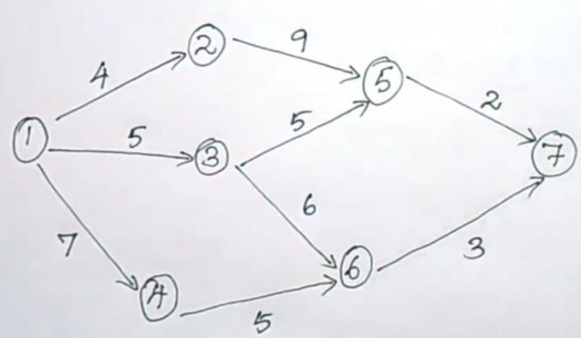
II Iteration



III Iteration



IV Iteration



1-2-5-7 = 4+9+2 = 15

1-3-5-7 = 5+5+2 = 12

1-3-6-7 = 5+6+3 = 14

1-4-6-7 = 7+5+3 = 15

Proj. comp. time = 15 weeks.

1-2-5-7 and 1-4-6-7

T.C= pre.total cost + direct cost(slope cost) - indirect cost

10,050 + 250 − 200

= 10,100

Final Result:

Since the total cost of this iteration (IV) is more than that of the previous iteration, stop the procedure and treat the solution of the previous iteration (III) as the best solution for implementation.

The final crashed project completion time is 16 weeks.

Corresponding critical paths are 1-2-5-7 and

1-4-6-7